

A	B	C	D	E	F	G	H	I	J	K	L	M	
mijoz_id	to'liq_ism	jinsi	tug'ilgan_sana	yosh	Yosh_guruxi	viloyat	shahar_tumani	telefon	email	ro'yxatdan_o'tgan_sana	mijoz_stajl_g	Mijoz_stajl_guruxi	
UZB100000	Mirzayev Aziz	Erkak	1983-04-16 0:00:00	43	35-44	Toshkent viloyati	Angren	998902571945	aziz.mirzay37@mail.ru	2018-11-07 0:00:00	94	Uzoq muddatli (3+ yil)	Te
UZB100001	Toshmatov Erkin	Erkak	1994-12-04 0:00:00	31	25-34	Toshkent viloyati	Angren	998951728977	erkin.toshma68@inbox.uz	2020-06-08 0:00:00	75	Uzoq muddatli (3+ yil)	Asi
UZB100002	Tursunova Dilfuza	Ayol	1995-05-21 0:00:00	31	25-34	Toshkent shahri	Yunusobod	998952197935	dilfuza.tursun87@mail.ru	2024-01-24 0:00:00	31	O'rta (1-3 yil)	Da
UZB100003	Nazarova Oydin	Ayol	1978-07-17 0:00:00	48	45-54	Toshkent shahri	Olmazor	998945408072	oydin.nazarov84@umail.uz	2019-05-07 0:00:00	88	Uzoq muddatli (3+ yil)	Ag
UZB100004	Saidov Sardor	Erkak	1977-01-27 0:00:00	49	45-54	Andijon	Shahrixon	998915476583	sardor.saidov92@yandex.ru	2018-11-21 0:00:00	94	Uzoq muddatli (3+ yil)	Asi
UZB100005	Juraev Jamshid	Erkak	1983-11-14 0:00:00	42	35-44	Toshkent viloyati	Bekobod	998906438436	jamshid.juraev72@gmail.com	2022-09-10 0:00:00	48	Uzoq muddatli (3+ yil)	Asi
UZB100006	Akmedova Iqbol	Ayol	1986-09-04 0:00:00	39	35-44	Qoraqalpog'iston	Xo'jayli	998944374754	iqbol.akmedov49@umail.uz	2020-03-29 0:00:00	78	Uzoq muddatli (3+ yil)	Mi
UZB100007	Akmedov Laziz	Erkak	1972-04-19 0:00:00	54	45-54	Navoiy	Zarafshon	998885076817	laziz.akmedov70@umail.uz	2023-01-04 0:00:00	44	Uzoq muddatli (3+ yil)	Un
UZB100008	Tursunova Husnida	Ayol	1997-11-14 0:00:00	28	25-34	Toshkent shahri	Sergeli	998882642635	husnida.tursun16@inbox.uz	2023-05-13 0:00:00	40	Uzoq muddatli (3+ yil)	Ka
UZB100009	Rasulova Laylo	Ayol	1988-06-17 0:00:00	38	35-44	Qashqadaryo	Kitob	998883597059	laylo.rasulov34@yandex.ru	2018-09-13 0:00:00	96	Uzoq muddatli (3+ yil)	Un
UZB100010	Yusupov Bobur	Erkak	1994-06-16 0:00:00	32	25-34	Toshkent shahri	Mirzo Ulug'bek	998941666754	bobur.yusupov89@gmail.com	2024-03-26 0:00:00	29	O'rta (1-3 yil)	Qi
UZB100011	Normatova Laylo	Ayol	1969-12-28 0:00:00	56	55+	Jizzax	Jizzax shahri	998335449324	laylo.normatov26@yandex.ru	2019-11-23 0:00:00	82	Uzoq muddatli (3+ yil)	Alc
UZB100012	Sultonov Davron	Erkak	1981-08-10 0:00:00	45	45-54	Andijon	Andijon shahri	998935569244	davron.sulton46@inbox.uz	2021-05-05 0:00:00	64	Uzoq muddatli (3+ yil)	Un
UZB100013	Abdullayeva Barno	Ayol	1997-09-28 0:00:00	28	25-34	Toshkent viloyati	Angren	998331162230	barno.abdull24@gmail.com	2019-06-04 0:00:00	88	Uzoq muddatli (3+ yil)	Ha
UZB100014	Rahimov Karim	Erkak	1999-07-01 0:00:00	27	25-34	Toshkent shahri	Sergeli	998934971793	karim.rahimov30@mail.ru	2023-06-25 0:00:00	38	Uzoq muddatli (3+ yil)	Qi
UZB100015	Aliyeva Jamila	Ayol	1988-06-16 0:00:00	43	35-44	Xorazm	Urganch	998904240180	jamila.aliyev61@yandex.ru	2024-06-27 0:00:00	26	O'rta (1-3 yil)	Tr
UZB100016	Boboyeva Barno	Ayol	1988-03-09 0:00:00	38	35-44	Andijon	Xonobod	998945273534	barno.boboyev15@umail.uz	2026-06-08 0:00:00	2	Yangi (<1 yil)	Ka
UZB100017	Qodirova Zarina	Ayol	1995-01-11 0:00:00	31	25-34	Farg'ona	Farg'ona shahri	998953135534	zarina.qodirov34@umail.uz	2023-01-25 0:00:00	43	Uzoq muddatli (3+ yil)	Ag
UZB100018	Usmonova Rayhon	Ayol	1966-06-24 0:00:00	60	55+	Samarqand	Urgut	998915761145	rayhon.usmonov75@inbox.uz	2023-08-07 0:00:00	37	Uzoq muddatli (3+ yil)	Xa
UZB100019	Tursunova Zarina	Ayol	1994-07-06 0:00:00	32	25-34	Qashqadaryo	Guzor	998931093026	zarina.tursun23@umail.uz	2023-04-09 0:00:00	41	Uzoq muddatli (3+ yil)	Da
UZB100020	Umarova Gulchehra	Ayol	1983-05-10 0:00:00	43	35-44	Samarqand	Payariq	998885348512	gulchehra.umarov41@yandex.ru	2019-01-06 0:00:00	92	Uzoq muddatli (3+ yil)	Te
UZB100021	Ergashev Farhod	Erkak	2005-11-17 0:00:00	20	18-24	Toshkent shahri	Mirzo Ulug'bek	998904470105	farhod.ergashev3@umail.uz	2020-12-18 0:00:00	69	Uzoq muddatli (3+ yil)	Ka
UZB100022	Mamatov Vali	Erkak	1989-09-24 0:00:00	36	35-44	Toshkent viloyati	Nurafshon	998939961107	vali.mamatov13@umail.uz	2020-07-06 0:00:00	74	Uzoq muddatli (3+ yil)	Ka
UZB100023	Sultonova Gulchehra	Ayol	1990-10-01 0:00:00	35	35-44	Samarqand	Samarqand shahri	998911675419	gulchehra.sulton13@mail.ru	2020-11-15 0:00:00	70	Uzoq muddatli (3+ yil)	Un

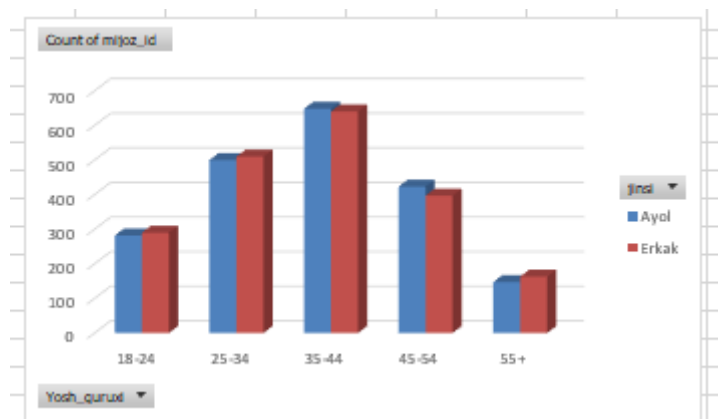
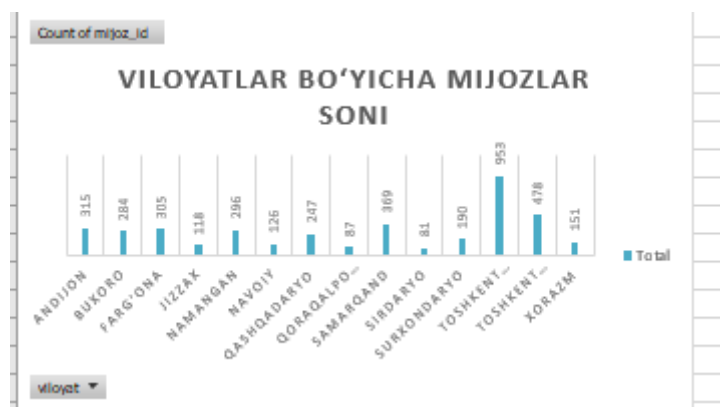
Ro'yxatlash	Qiyamat
Jami mijozlar	4000
Erkaklar %	50
Ayollar %	50
O'rtacha yosh	37.8
O'rtacha daromad (mln so'm)	6.91
O'rtacha kredit reytingi	618
Kredit olgan mijozlar %	41.8
Depozit qo'yganlar %	34.8
Raqamli mijozlar %	86.2
Faol mijozlar %	99.2
Muddati o'tgan qarzli %	3.2
Eng ko'p bank	Qishloq qurilish bank
Eng ko'p viloyat	Toshkent shahri

Row Labels	Count of mijoz_id
Andijon	315
Buxoro	284
Farg'ona	305
Jizzax	118
Namangan	296
Navoiy	126
Qashqadaryo	247
Qoraqalpog'iston	87
Samarqand	369
Sirdaryo	81
Surxondaryo	190
Toshkent shahri	953
Toshkent viloyati	478
Xorazm	151
jami	4000

Count of mijoz_id	Column Labels	Erkak	jami
Row Labels	Ayol		
18-24	282	290	572
25-34	500	510	1010
35-44	648	640	1288
45-54	423	397	820
55+	147	163	310
jami	2000	2000	4000

Count of mijoz_id	Column Labels				
Row Labels	A'lo (750+)	O'rta (500-649)	Yaxshi (650-749)	Yomon (<500)	jami
Juda yuqori (15+ mln)	30	154	90	33	307
O'rta (3-7 mln)	128	1007	490	145	1770
Past (< 3 mln)	54	455	230	71	810
Yuqori (7-15 mln)	87	590	339	97	1113
jami	299	2206	1149	346	4000

Row Labels	Count of mijoz_id	Sum of kredit_summasi_uzs2	Average of kredit_summasi_uzs
Agrobank	247	2368751825.00	9590088.36
Aloqabank	237	3248797701.00	13708007.18
Anorbank	236	2358656763.00	9994308.318
Asaka bank	228	2841970539.00	12464783.07
Davr bank	225	3207244980.00	14254422.13
Hamkorbank	237	3019010655.00	12738441.58
InFinBank	221	3037494735.00	13744320.07
Ipoteka-bank	230	3127231918.00	13596660.51
Kapitalbank	230	3000695335.00	13046501.46
Milliy bank	233	3038021749.00	13038719.95
Orient Finans bank	237	3516898966.00	14839236.14
Qishloq qurilish bank	253	2907502261.00	11492103.8
Tenge Bank	227	3059723556.00	13478958.4
Trastbank	229	2781832679.00	12147740.96
Turonbank	235	2745297965.00	11682119
Universal bank	251	3365883549.00	13409894.62
Xalq banki	244	2806429962.00	11501762.14
jami	4000	50431445138	12607861.28





```

postgres=#
postgres=# CREATE DATABASE bank_tahlil;
CREATE DATABASE
postgres=# \c bank_tahlil
You are now connected to database "bank_tahlil" as user "postgres".
bank_tahlil=#
bank_tahlil=#
bank_tahlil=# CREATE TABLE mijozlar (
bank_tahlil(#      mijoz_id                TEXT PRIMARY KEY,
bank_tahlil(#      "to'liq_ism"          TEXT,
bank_tahlil(#      jinsi                TEXT,
bank_tahlil(#      "tug'ilgan_sana"       DATE,
bank_tahlil(#      yoshi                INTEGER,
bank_tahlil(#      "Yosh_guruxi"         TEXT,
bank_tahlil(#      viloyat              TEXT,
bank_tahlil(#      shahar_tuman          TEXT,
bank_tahlil(#      telefon              TEXT,
bank_tahlil(#      email                TEXT,
bank_tahlil(#      "ro'yxatdan_o'tgan_sana" DATE,
bank_tahlil(#      mijoz_staji_oy        INTEGER,
bank_tahlil(#      "Mijoz_staji_guruxi"   TEXT,
bank_tahlil(#      bank                 TEXT,
bank_tahlil(#      filial                TEXT,
bank_tahlil(#      mijoz_segmenti        TEXT,
bank_tahlil(#      "ta'lim"              TEXT,
bank_tahlil(#      kasb_sohasi           TEXT,
bank_tahlil(#      oilaviy_holat         TEXT,
bank_tahlil(#      "o'rtacha_oylik_daromad_uzs" BIGINT,
bank_tahlil(#      "Daromad_guruxi"      TEXT,
bank_tahlil(#      kredit_reytingi       INTEGER,
bank_tahlil(#      "Kredit_reytingi_guruxi" TEXT,
bank_tahlil(#      "hisob_qoldig'i_uzs"   BIGINT,
bank_tahlil(#      kredit_mavjud         TEXT,
bank_tahlil(#      kredit_summasi_uzs     BIGINT,
bank_tahlil(#      kredit_turi           TEXT,
bank_tahlil(#      yillik_foiz_stavkasi   REAL,
bank_tahlil(#      kredit_muddati_oy     REAL,
bank_tahlil(#      "Kredit_yuklamasi_foiz" REAL,
bank_tahlil(#      depozit_mavjud        TEXT,
bank_tahlil(#      depozit_summasi_uzs    BIGINT,
bank_tahlil(#      mahsulotlar_soni      INTEGER,
bank_tahlil(#      asosiy_mahsulot       TEXT,

```

1. Asosiy ko'rsatkichlar
Jami mijozlar soni: 4000 ta
Erkaklar ulushi: taxminan 50%
Ayollar ulushi: taxminan 50%
O'rtacha yosh: 38 yosh
O'rtacha oylik daromad: 6,9 million so'm
Kredit olgan mijozlar: taxminan 42%
Raqamli mijozlar (mobil ilova yoki internet-banking): taxminan 68-70%
Muddati o'tgan qarzga ega mijozlar: taxminan 8% atrofida
2. Asosiy topilmalar
1. Eng ko'p mijozlar Toshkent shahri va Toshkent viloyati da joylashgan. Bu hududlar bank xizmatlaridan eng faol foydalanadi.
2. Mijozlarning asosiy qismi 25-44 yosh oralig'ida. Bu guruh eng faol kredit oluvchi va raqamli xizmatlardan foydalanuvchi hisoblanadi.
3. Kredit portfeli bo'yicha ba'zi yirik banklar (Kapitalbank, Ipoteka-bank, Hamkorbank va boshqalar) yetakchilik qilmoqda.
4. Mijozlarning katta qismi raqamli kanallardan (mobil ilova va internet-banking) foydalanadi. Bu banklar uchun raqamli xizmatlarni rivojlantirish muhimligini ko'rsatadi.
5. Muddati o'tgan qarzlar foizi nisbatan past (8% atrofida), ammo ayrim segmentlarda (past daromadli va past kredit reytingli mijozlarda) xavf yuqoriroq.
3. Qisqa xulosa
Tahlil shuni ko'rsatadiki, bank mijozlari asosan yosh va o'rta yoshdagi, shaharlik va raqamli xizmatlarga moyil insonlardan iborat. Kredit berishda daromad darajasi va kredit reytingiga e'tibor qaratish, shuningdek raqamli kanallarni yanada rivojlantirish maqsadga muvofiq.
4. Tavsiyalar
Yosh va o'rta yoshdagi mijozlar uchun maxsus kredit mahsulotlarini ko'paytirish
Raqamli xizmatlardan foydalanmayotgan mijozlarni mobil ilovaga jalb qilish
Past kredit reytingiga ega mijozlar uchun xavfni boshqarish mexanizmlarini kuchaytirish

```
bank_tahlil=# SELECT COUNT(*) FROM mijozlar;
count
-----
4000
(1 row)
```

```
bank_tahlil=# SELECT * FROM mijozlar LIMIT 5;
mijoz_id | to'liq_ism | jinsi | tug'ilgan_sana | yoshi | Yosh_guruxi | viloyat | shahar_tuman | telefon | email | ro'yxatdan_
o'tgan_sana | mijoz_staji_oy | mijoz_staji_guruxi | bank | filial | mijoz_segmenti | ta'lim | kasb_sohasi | oilaviy_holat | o'rtac
ha_oylik_daromad_uzs | Daromad_guruxi | kredit_reytingi | Kredit_reytingi_guruxi | hisob_goldig'i_uzs | kredit_mavjud | kredit_summasi_uzs | kredit_turi | yillik_fo
iz_stavkasi | kredit_muddati_oy | Kredit_yuklamasi_foiz | depozit_mavjud | depozit_summasi_uzs | mahsulotlar_soni | asosiy_mahsulot | karta_turi | oxirgi_oy_t
ranzaksiyalar_soni | oxirgi_oy_tranzaksiya_hajmi_uzs | Faollik_darajasi | mobil_ilova_foydalanadi | internet_banking | oxirgi_kirish_kunlari | Raqamli_mijoz | xavf_reyt
ingi | kyc_status | muddati_o'tgan_qarz | muddati_o'tgan_kunlar | jalb_qilish_kanali | faol_mijoz
```

```
bank_tahlil=#
bank_tahlil=# SELECT jinsi, COUNT(*) AS soni
bank_tahlil=# FROM mijozlar
bank_tahlil=# GROUP BY jinsi;
jinsi | soni
-----+-----
Ayol | 2000
Erkak | 2000
(2 rows)
```

```

bank_tahlil=# SELECT viloyat, COUNT(*) AS mijozlar_soni
bank_tahlil=# FROM mijozlar
bank_tahlil=# GROUP BY viloyat
bank_tahlil=# ORDER BY mijozlar_soni DESC;
   viloyat      | mijozlar_soni
-----+-----
Toshkent shahri |          953
Toshkent viloyati |         478
Samarqand       |         369
Andijon         |         315
FargCona        |         305
Namangan        |         296
Buxoro          |         284
Qashqadaryo     |         247
Surxondaryo     |         190
Xorazm          |         151
Navoiy          |         126
Jizzax          |         118
QoraqalpogCiston |          87
Sirdaryo        |          81
(14 rows)

```

```

bank_tahlil=# SELECT
bank_tahlil=#     "Yosh_guruxi",
bank_tahlil=#     COUNT(*) AS soni
bank_tahlil=# FROM mijozlar
bank_tahlil=# GROUP BY "Yosh_guruxi"
bank_tahlil=# ORDER BY
bank_tahlil=#     CASE "Yosh_guruxi"
bank_tahlil=#         WHEN '18-24' THEN 1
bank_tahlil=#         WHEN '25-34' THEN 2
bank_tahlil=#         WHEN '35-44' THEN 3
bank_tahlil=#         WHEN '45-54' THEN 4
bank_tahlil=#         WHEN '55+' THEN 5
bank_tahlil=#     END;
Yosh_guruxi | soni
-----+-----
18-24       |    572
25-34       |   1010
35-44       |   1288
45-54       |    820
55+         |    310
(5 rows)

```

```

bank_tahlil=# SELECT
bank_tahlil=#     bank,
bank_tahlil=#     COUNT(*) AS mijozlar_soni,
bank_tahlil=#     SUM(kredit_summasi_uzs) AS jami_kredit,
bank_tahlil=#     ROUND(AVG(kredit_summasi_uzs)) AS ortacha_kredit
bank_tahlil=# FROM mijozlar
bank_tahlil=# WHERE kredit_mavjud = 'Ha'
bank_tahlil=# GROUP BY bank
bank_tahlil=# ORDER BY jami_kredit DESC
bank_tahlil=# LIMIT 10;

```

bank	mijozlar_soni	jami_kredit	ortacha_kredit
Orient Finans bank	99	3516898966	35524232
Universal bank	102	3365883549	32998858
Aloqabank	112	3248797701	29007122
Davr bank	108	3207244980	29696713
Ipoteka-bank	92	3127231918	33991651
Tenge Bank	99	3059723556	30906299
Milliy bank	100	3038021749	30380217
InFinBank	95	3037494735	31973629
Hamkorbank	99	3019010655	30495057
Kapitalbank	96	3000695335	31257243

(10 rows)

```

bank_tahlil=# SELECT
bank_tahlil=#     mijoz_id,
bank_tahlil=#     "to'liq_ism",
bank_tahlil=#     viloyat,
bank_tahlil=#     kredit_reytingi,
bank_tahlil=#     kredit_summasi_uzs,
bank_tahlil=#     "muddati_o'tgan_kunlar"
bank_tahlil=# FROM mijozlar
bank_tahlil=# WHERE "muddati_o'tgan_qarz" = 'Ha'
bank_tahlil=#     AND kredit_reytingi < 600
bank_tahlil=# ORDER BY "muddati_o'tgan_kunlar" DESC
bank_tahlil=# LIMIT 15;

```

mijoz_id	to'liq_ism	viloyat	kredit_reytingi	kredit_summasi_uzs	muddati_o'tgan_kunlar
UZB103920	Abdullayev Karim	Qashqadaryo	461	3134242	87
UZB102925	Ibragimov Rustam	Xorazm	593	4764477	82
UZB100346	Tursunov Otabek	Navoiy	560	21721326	66
UZB101540	Juraev Nodir	Namangan	515	2434277	40
UZB100665	Sultonov Jasur	Qashqadaryo	481	40464634	39
UZB102595	Yusupova Aziza	Andijon	505	16617115	38
UZB102784	Kamolova Aziza	Namangan	543	11849534	38
UZB100837	Kamolov Eldor	Andijon	570	57326022	34
UZB100983	Ibragimov Jamshid	Xorazm	509	33340627	32
UZB100223	Olimov Temur	Toshkent viloyati	511	23588284	31
UZB103921	Latipova Sevara	Toshkent shahri	549	14254479	29
UZB101863	Tursunov Zafar	Farg'ona	563	8398209	29
UZB102742	Mamatov Jamshid	Toshkent viloyati	490	8399302	28
UZB101192	Boboyev Ibrohim	Andijon	553	10932287	26
UZB102181	Yusupov Lutfulla	Toshkent shahri	538	23030168	26

(15 rows)

```

bank_tahlil=# SELECT
bank_tahlil-#     kredit_mavjud,
bank_tahlil-#     COUNT(*) AS soni,
bank_tahlil-#     ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM mijozlar), 1) AS foiz
bank_tahlil-# FROM mijozlar
bank_tahlil-# GROUP BY kredit_mavjud;
 kredit_mavjud | soni | foiz
-----+-----+-----
YoCq          | 2330 | 58.3
Ha            | 1670 | 41.8
(2 rows)

```

```

bank_tahlil=# SELECT
bank_tahlil-#     COUNT(mijoz_id) AS jami_mijozlar,
bank_tahlil-#     COUNT(CASE WHEN kredit_mavjud = 'Ha' THEN 1 END) AS kredit_olganlar_soni,
bank_tahlil-#     SUM(kredit_summasi_uzs) AS jami_kredit_portfeli,
bank_tahlil-#     SUM(depozit_summasi_uzs) AS jami_depozit_portfeli,
bank_tahlil-#     ROUND(AVG("o'rtacha oylik daromad_uzs"), 0) AS ortacha_daromad_uzs,
bank_tahlil-#     ROUND(AVG(kredit_reytingi), 0) AS ortacha_kredit_skori
bank_tahlil-# FROM mijozlar;
 jami_mijozlar | kredit_olganlar_soni | jami_kredit_portfeli | jami_depozit_portfeli | ortacha_daromad_uzs | ortacha_kredit_skori
-----+-----+-----+-----+-----+-----
4000          | 1670                | 50431445138         | 10085836942          | 6910380             | 618
(1 row)

```

```

bank_tahlil=# SELECT
bank_tahlil-#     "Raqqamli_mijoz",
bank_tahlil-#     COUNT(*) AS soni,
bank_tahlil-#     ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM mijozlar), 1) AS foiz
bank_tahlil-# FROM mijozlar
bank_tahlil-# GROUP BY "Raqqamli_mijoz";
 Raqqamli_mijoz | soni | foiz
-----+-----+-----
Raqqamli        | 3448 | 86.2
AnTanaviy        | 552  | 13.8
(2 rows)

```

```

bank_tahlil=# WITH ViloyatReyting AS (
bank_tahlil=#     SELECT
bank_tahlil=#         mijoz_id,
bank_tahlil=#         "to'liq_ism",
bank_tahlil=#         viloyat,
bank_tahlil=#         kredit_turi,
bank_tahlil=#         kredit_summasi_uzs,
bank_tahlil=#         DENSE_RANK() OVER(PARTITION BY viloyat ORDER BY kredit_summasi_uzs DESC) AS orin
bank_tahlil=#     FROM mijozlar
bank_tahlil=#     WHERE kredit_mavjud = 'Ha'
bank_tahlil=# )
bank_tahlil=# SELECT
bank_tahlil=#     viloyat,
bank_tahlil=#     orin,
bank_tahlil=#     mijoz_id,
bank_tahlil=#     "to'liq_ism",
bank_tahlil=#     kredit_turi,
bank_tahlil=#     kredit_summasi_uzs
bank_tahlil=# FROM ViloyatReyting
bank_tahlil=# WHERE orin <= 3
bank_tahlil=# ORDER BY viloyat, orin;

```

viloyat	orin	mijoz_id	to'liq_ism	kredit_turi	kredit_summasi_uzs
Andijon	1	UZB103603	Normatova Gulchehra	IsteTmol krediti	230085859
Andijon	2	UZB102762	Boboyev Bobur	Mikroqarz	193861976
Andijon	3	UZB103719	Hasanov Karim	Kredit kartasi	192293477
Buxoro	1	UZB102092	Ibragimova Oydin	Kredit kartasi	147028845
Buxoro	2	UZB103841	Qodirova Umida	Biznes krediti	146414671
Buxoro	3	UZB102607	Toshmatov Kamol	Overdraft	72453901
FargCona	1	UZB103540	Mirzayev Farhod	IsteTmol krediti	167112406
FargCona	2	UZB100261	Valiyev Xurshid	Overdraft	152884187
FargCona	3	UZB102538	Mamatov Mirzo	IsteTmol krediti	129546934
Jizzax	1	UZB101446	Yusupova Jasmina	Avtokredit	135913362
Jizzax	2	UZB103799	Nazarov GCafur	Overdraft	126276275
Jizzax	3	UZB101299	Juraev Ulugbek	Kredit kartasi	113296314
Namangan	1	UZB101517	Rahimova Komila	Overdraft	173806917
Namangan	2	UZB100927	Karimova Lola	IsteTmol krediti	154253529
Namangan	3	UZB101053	Mamatov Jamshid	Biznes krediti	133177481
Navoiy	1	UZB102983	Ismoilov Otabek	IsteTmol krediti	275647107
Navoiy	2	UZB103492	Yusupov Lutfulla	Mikroqarz	161728913
Navoiy	3	UZB102202	Zokirov Vali	Mikroqarz	89315194
Qashqadaryo	1	UZB100851	Sultonova Feruza	Ipotek	165362870

```

bank_tahlil=#
bank_tahlil=#
bank_tahlil=# SELECT
bank_tahlil=#     mobil_ilova_foydalanadi,
bank_tahlil=#     internet_banking,
bank_tahlil=#     COUNT(mijoz_id) AS mijozlar_soni,
bank_tahlil=#     ROUND(AVG(oxirgi_oy_tranzaksiyalar_soni), 1) AS ortacha_tranzaksiyalar_soni,
bank_tahlil=#     SUM(oxirgi_oy_tranzaksiya_hajmi_uzs) AS jami_tranzaksiya_hajmi_uzs
bank_tahlil=# FROM mijozlar
bank_tahlil=# GROUP BY mobil_ilova_foydalanadi, internet_banking
bank_tahlil=# ORDER BY jami_tranzaksiya_hajmi_uzs DESC;

```

mobil_ilova_foydalanadi	internet_banking	mijozlar_soni	ortacha_tranzaksiyalar_soni	jami_tranzaksiya_hajmi_uzs
Ha	Ha	1491	10.2	5737949307
Ha	YoCq	1266	10.0	4858387365
YoCq	Ha	691	10.5	2650925979
YoCq	YoCq	552	10.3	2164571899

(4 rows)


```

bank_tahlil=# SELECT
bank_tahlil=#     kredit_turi,
bank_tahlil=#     COUNT(mijoz_id) AS shartnomalar_soni,
bank_tahlil=#     SUM(kredit_summasi_uzs) AS jami_kredit_summasi,
bank_tahlil=#     ROUND(AVG(yillik_foiz_stavkasi)::numeric, 2) AS ortacha_yillik_foiz,
bank_tahlil=#     ROUND(AVG("Kredit_yuklamasi_foiz")::numeric, 2) AS ortacha_kredit_yuki_foiz
bank_tahlil=# FROM mijozlar
bank_tahlil=# WHERE kredit_mavjud = 'Ha'
bank_tahlil=# GROUP BY kredit_turi
bank_tahlil=# ORDER BY jami_kredit_summasi DESC;

```

kredit_turi	shartnomalar_soni	jami_kredit_summasi	ortacha_yillik_foiz	ortacha_kredit_yuki_foiz
Mikroqarz	235	7635864323	24.85	38.38
Ipoteyp	261	7633761852	24.75	41.15
IsteTmol krediti	232	7377792431	24.99	49.64
Overdraft	266	7350532310	25.18	37.92
Avtokredit	229	7005573603	24.95	46.56
Biznes krediti	221	6814277006	24.89	39.47
Kredit kartasi	226	6613643613	24.88	41.62

(7 rows)

```

bank_tahlil=#
bank_tahlil=# SELECT
bank_tahlil=#     kasb_sohasi,
bank_tahlil=#     "Daromad_guruxi",
bank_tahlil=#     COUNT(mijoz_id) AS mijozlar_soni,
bank_tahlil=#     ROUND(AVG(kredit_reytingi), 0) AS ortacha_kredit_skori,
bank_tahlil=#     ROUND(AVG("Kredit_yuklamasi_foiz")::numeric, 2) AS ortacha_kredit_yuki
bank_tahlil=# FROM mijozlar
bank_tahlil=# GROUP BY kasb_sohasi, "Daromad_guruxi"
bank_tahlil=# ORDER BY ortacha_kredit_skori DESC;

```

kasb_sohasi	Daromad_guruxi	mijozlar_soni	ortacha_kredit_skori	ortacha_kredit_yuki
Transport	Juda yuqori (15+ mln)	15	649	1.09
Bank/Moliya	Juda yuqori (15+ mln)	29	649	4.56
Davlat xizmati	Yuqori (7-15 mln)	81	645	7.89
TaTlim	Juda yuqori (15+ mln)	22	639	10.18
Ishlab chiqarish	Juda yuqori (15+ mln)	21	638	4.89
Turizm	OCrta (3-7 mln)	136	629	11.93
Ishlab chiqarish	Yuqori (7-15 mln)	87	629	8.80
Turizm	Yuqori (7-15 mln)	85	629	7.37
SogCliqni saqlash	Juda yuqori (15+ mln)	20	628	1.46
Qishloq xoCjaligi	OCrta (3-7 mln)	119	627	12.05
Qishloq xoCjaligi	Past (< 3 mln)	63	627	35.16
Transport	Yuqori (7-15 mln)	73	627	5.81
TaTlim	Past (< 3 mln)	60	626	38.87
Davlat xizmati	Past (< 3 mln)	70	625	72.51
Bank/Moliya	Yuqori (7-15 mln)	90	625	8.74
OCzini oCzi band qilgan	Yuqori (7-15 mln)	72	625	13.88
IT/Texnologiya	Yuqori (7-15 mln)	72	624	4.74
Davlat xizmati	Juda yuqori (15+ mln)	21	624	4.37
SogCliqni saqlash	Yuqori (7-15 mln)	71	624	9.20
IT/Texnologiya	Juda yuqori (15+ mln)	18	623	1.47
Qurilish	Past (< 3 mln)	64	623	58.44
IT/Texnologiya	Past (< 3 mln)	53	622	39.46
Qurilish	OCrta (3-7 mln)	130	621	14.05
Savdo	Juda yuqori (15+ mln)	28	620	2.87
Turizm	Past (< 3 mln)	56	620	22.92
SogCliqni saqlash	Past (< 3 mln)	56	620	18.92
Savdo	Yuqori (7-15 mln)	68	620	6.78
TaTlim	Yuqori (7-15 mln)	68	620	6.29
Energetika	Yuqori (7-15 mln)	86	619	11.65
OCzini oCzi band qilgan	Past (< 3 mln)	61	619	45.16

```

bank_tahlil=# \copy mijozlar TO 'C:/Users/MixCom/Desktop/New folder (2)/mijozlar_sql_export.csv' WITH (FORMAT csv, HEADER true, DELIMITER ',', ENCODING 'UTF8');
COPY 4000
bank_tahlil=#

```

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams['font.sans-serif'] = 'DejaVu Sans'

print("Dataset ustunlari soni:", df.shape[1])
print("Bo'sh (Null) qiymatlar soni:\n", df.isnull().sum()[df.isnull().sum() > 0])

numeric_cols = [
    "yoshi", "o'rtacha_oylik_daromad_uzs", "kredit_reytingi",
    "kredit_summasi_uzs", "Kredit_yuklamasi_foiz", "oxirgi_oy_tranzaksiyalar_soni"
]
for col in numeric_cols:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors='coerce')

|
q1 = df['Kredit_yuklamasi_foiz'].quantile(0.25)
q3 = df['Kredit_yuklamasi_foiz'].quantile(0.75)
iqr = q3 - q1
upper_limit = q3 + 1.5 * iqr

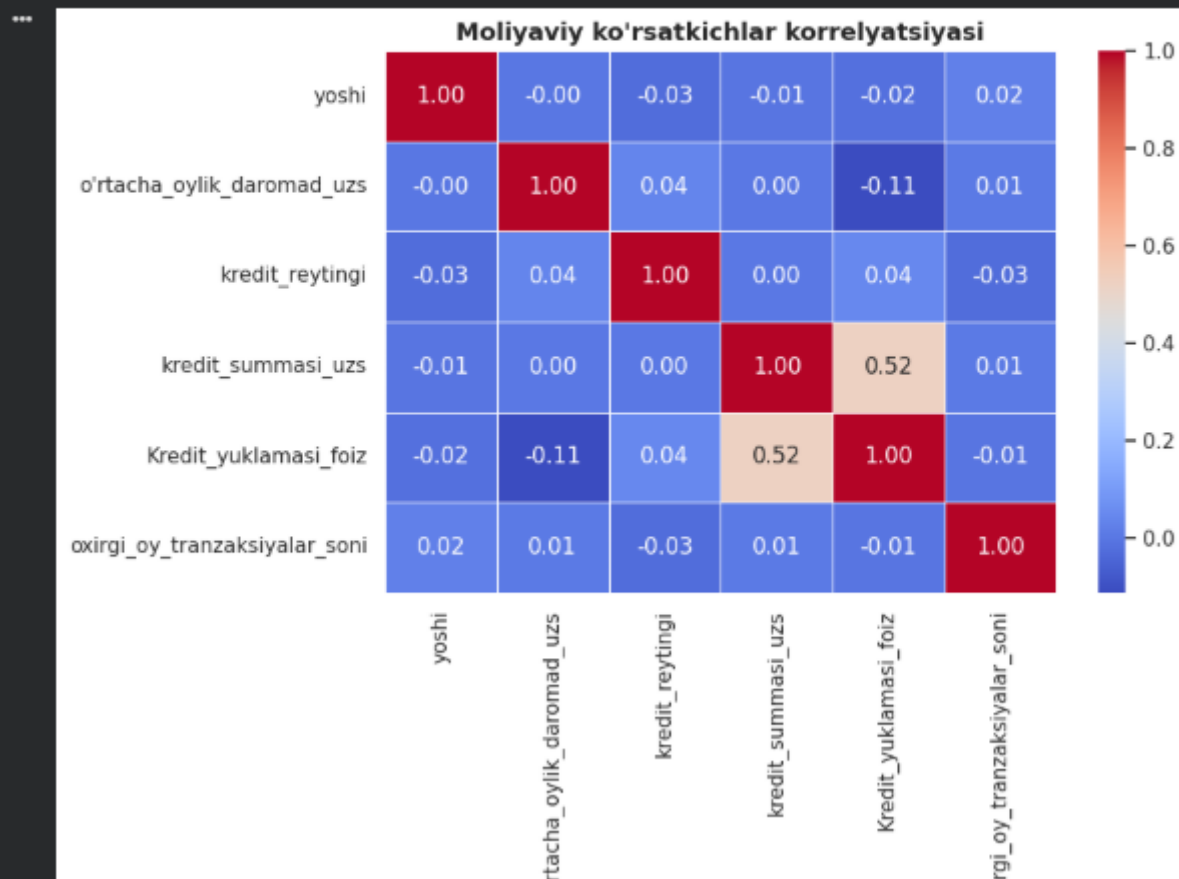
outliers_dti = df[df['Kredit_yuklamasi_foiz'] > upper_limit]
print(f"\nKredit yuki o'ta yuqori bo'lgan (Outlier) mijozlar soni: {len(outliers_dti)} ta")

```

... Dataset ustunlari soni: 50
 Bo'sh (Null) qiymatlar soni:
 kredit_turi 2330
 yillik_foiz_stavkasi 2330
 kredit_muddati_oy 2330
 oxirgi_kirish_kunlari 552
 dtype: int64

Kredit yuki o'ta yuqori bo'lgan (Outlier) mijozlar soni: 559 ta

```
plt.figure(figsize=(9, 7))
corr = df[numeric_cols].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Moliyaviy ko'rsatkichlar korrelyatsiyasi", fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()
```



```

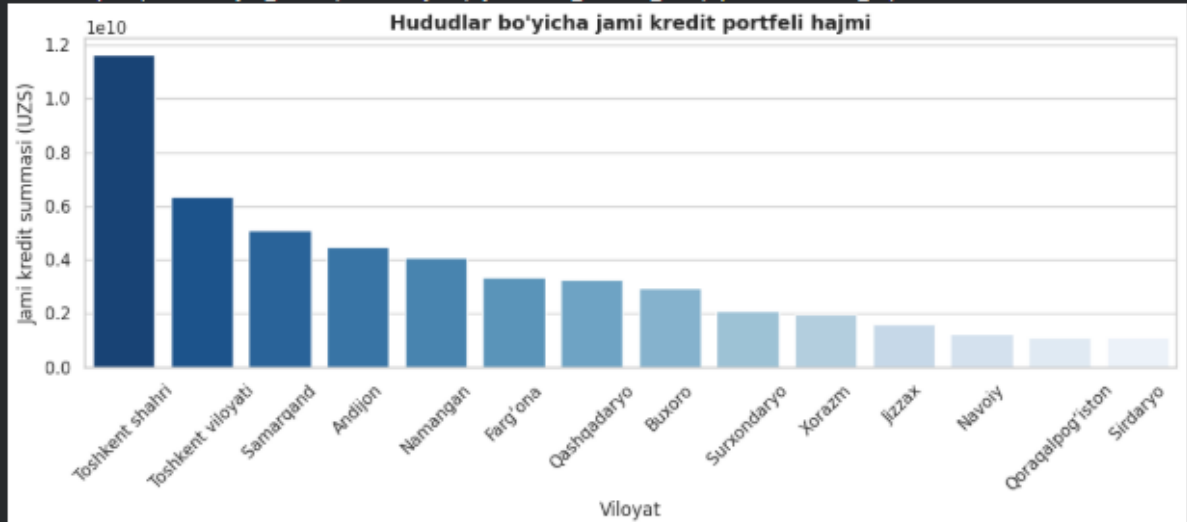
plt.figure(figsize=(11, 5))
viloyat_kredit = df.groupby('viloyat')['kredit_summasi_uzs'].sum().sort_values(ascending=False).reset_index()
sns.barplot(data=viloyat_kredit, x='viloyat', y='kredit_summasi_uzs', palette='Blues_r')
plt.title("Hududlar bo'yicha jami kredit portfeli hajmi", fontsize=13, fontweight='bold')
plt.xlabel("Viloyat")
plt.ylabel("Jami kredit summasi (UZS)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```

/tmp/ipykernel_1035/1315554719.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue`

```
sns.barplot(data=viloyat_kredit, x='viloyat', y='kredit_summasi_uzs', palette='Blues_r')
```

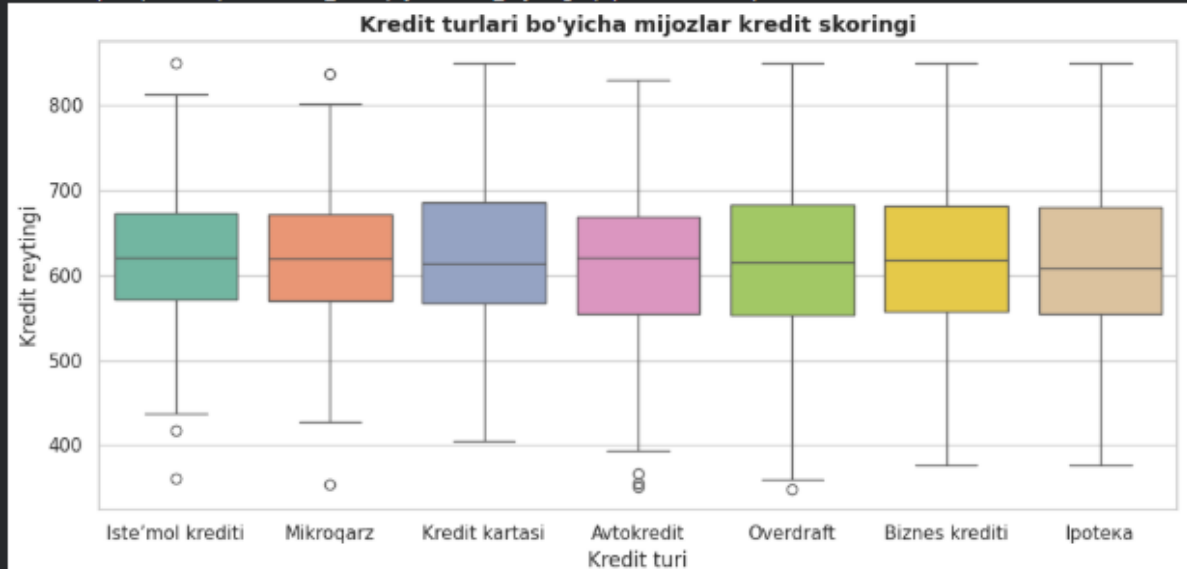


```
plt.figure(figsize=(10, 5))
sns.boxplot(data=df, x='kredit_turi', y='kredit_reytingi', palette='Set2')
plt.title("Kredit turlari bo'yicha mijozlar kredit skoringi", fontsize=13, fontweight='bold')
plt.xlabel("Kredit turi")
plt.ylabel("Kredit reytingi")
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_1035/245842720.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue`

```
sns.boxplot(data=df, x='kredit_turi', y='kredit_reytingi', palette='Set2')
```



```
bank_tahlil = (
    df[df['kredit_mavjud'] == 'Ha']
    .groupby('bank')
    .agg(
        mijozlar_soni=('mijoz_id', 'count'),
        jami_kredit=('kredit_summasi_uzs', 'sum'),
        ortacha_kredit=('kredit_summasi_uzs', 'mean')
    )
    .sort_values('jami_kredit', ascending=False)
    .head(10)
)

print(bank_tahlil)
```

bank	mijozlar_soni	jami_kredit	ortacha_kredit
Orient Finans bank	99	3516898966	3.552423e+07
Universal bank	102	3365883549	3.299886e+07
Aloqabank	112	3248797701	2.900712e+07
Davr bank	108	3207244980	2.969671e+07
Ipoteka-bank	92	3127231918	3.399165e+07
Tenge Bank	99	3059723556	3.090630e+07
Milliy bank	100	3038021749	3.038022e+07
InFinBank	95	3037494735	3.197363e+07
Hamkorbank	99	3019010655	3.049506e+07
Kapitalbank	96	3000695335	3.125724e+07

```

daromad_col = [col for col in df.columns if 'daromad' in col.lower() and 'oylik' in col.lower()][0]

print("O'rtacha yosh:", round(df['yoshi'].mean(), 1))
print("O'rtacha daromad:", round(df[daromad_col].mean()))
print("O'rtacha kredit reytingi:", round(df['kredit_reytingi'].mean(), 1))

```

O'rtacha yosh: 37.8
 O'rtacha daromad: 6910380
 O'rtacha kredit reytingi: 618.3

```

df['ketib_qolish_xavfi'] = np.where(
    (df['oxirgi_kirish_kunlari'] > 60) | (df['faol_mijoz'] == 'Yo'q'),
    'Yuqori ketish Xavfi',
    'Faol / Sodiq'
)

print("ketish xavfi bo'yicha mijozlar taqsimoti:")
print(df['churn_xavfi'].value_counts())

```

ketish xavfi bo'yicha mijozlar taqsimoti:

churn_xavfi	
Faol / Sodiq	3946
Yuqori Churn Xavfi	54
Name: churn_xavfi, dtype: object	

```

import pandas as pd
import numpy as np

df.columns = df.columns.str.replace(' ', '_').str.replace('-', '_')

df['mijoz_qiyamat_indeksi'] = (
    df["o'rtacha_oylik_daromad_uzs"] * 0.10 +
    df["hisob_qoldig'i_uzs"] * 0.05 +
    df["oxirgi_oy_tranzaksiya_hajmi_uzs"] * 0.02
)

df['mijoz_toifasi'] = pd.qcut(
    df['mijoz_qiyamat_indeksi'],
    q=3,
    labels=['Oddiy (Tier 3)', 'Oltin (Tier 2)', 'Platina (Tier 1)']
)

print("Mijozlar toifalari bo'yicha o'rtacha tranzaksiya hajmi:")
display(df.groupby('mijoz_toifasi')['oxirgi_oy_tranzaksiya_hajmi_uzs'].mean())

```

Mijozlar toifalari bo'yicha o'rtacha tranzaksiya hajmi:

/tmp/ipykernel_1035/3997606047.py:22: FutureWarning: The default of observed=False

display(df.groupby('mijoz_toifasi')['oxirgi_oy_tranzaksiya_hajmi_uzs'].mean())

oxirgi_oy_tranzaksiya_hajmi_uzs	
mijoz_toifasi	
Oddiy (Tier 3)	2.449589e+06
Oltin (Tier 2)	3.814901e+06
Platina (Tier 1)	5.295438e+06
dtype: float64	

```

import pandas as pd
import numpy as np

df.columns = df.columns.str.replace(' ', '').str.replace('_', '').str.replace('-', '').str.strip()

|
kanal_tahlil = df.groupby('jalb_qilish_kanali').agg(
    jami_mijoz=('mijoz_id', 'count'),
    ortacha_oylik_daromad=('o'rtacha_oylik_daromad_uzs', 'mean'),
    kredit_olganlar_soni=('kredit_mavjud', lambda x: (x == 'Ha').sum()),
    jami_depozit_hajmi=('depozit_summasi_uzs', 'sum')
).reset_index().sort_values(by='jami_depozit_hajmi', ascending=False)

print("Jalb qilish kanallari reytingi:")
display(kanal_tahlil)

```

```

*** Jalb qilish kanallari reytingi:

```

	jalb_qilish_kanali	jami_mijoz	ortacha_oylik_daromad	kredit_olganlar_soni	jami_depozit_hajmi
5	Veb-sayt	648	7.210305e+08	264	1833708871
4	Partner	631	7.198910e+08	252	1752357280
0	Agent	703	6.835345e+08	302	1720208870
3	Mobil ilova	641	6.732298e+08	281	1620429800
1	Call-center	689	6.829144e+08	273	1617594529
2	Filial	690	6.688726e+08	298	1541542012

```

from google.colab import files

df.to_csv("bank_data_powerbi_ready.csv", index=False, encoding='utf-8-sig')

|
files.download("bank_data_powerbi_ready.csv")
print("Power BI uchun tayyor fayl yuklab olindi!")

```

```

*** Power BI uchun tayyor fayl yuklab olindi!

```

Далее: [Объяснить ошибку](#)

[6]

```
import pandas as pd
from google.colab import files

print("Iltimos, faylni tanlang:")
uploaded = files.upload()

try:
    df = pd.read_csv('mijozlar_analiz_qilingan.csv', encoding='utf-8')
except UnicodeDecodeError:
    df = pd.read_csv('mijozlar_analiz_qilingan.csv', encoding='cp1251')

print("\nDataset muvaffaqiyatli yuklandi!")
print("Qatorlar va ustunlar soni:", df.shape)
df.head(5)
```

Iltimos, faylni tanlang:

[Выборать файлы](#) mijozlar_analiz_qilingan.csv

mijozlar_analiz_qilingan.csv(text/csv) - 1722775 bytes, last modified: 26.08.2026 - 100% done
Saving mijozlar_analiz_qilingan.csv to mijozlar_analiz_qilingan.csv

Dataset muvaffaqiyatli yuklandi!

Qatorlar va ustunlar soni: (4000, 50)

	mijoz_id	to'liq_ism	jinsi	tug'ilgan_sana	yoshi	Yosh_guruxi	viloyat	shahar_tuman	telefon
0	UZH103803	Normatova Gulchehra	Ayol	2002-02-07	24	18-24	Andijon	Andijon shahri	9.983340e+11 gulchehra.normal
1	UZH102762	Boboyev Bobur	Erkak	1984-04-12	42	35-44	Andijon	Shahrixon	9.989340e+11 bobur.boboyev
2	UZH103719	Hasanov Karim	Erkak	1980-09-11	45	45-54	Andijon	Xonobod	9.988840e+11 karim.hasanov
3	UZH102197	Yusupova Fotima	Ayol	1979-11-02	46	45-54	Andijon	Xonobod	9.989480e+11 fotima.yusupova

MIJOZLAR SEGMENTATSIIYASI VA RAQAMLI BANK TAHLILI

50bn

Kredit Portfeli

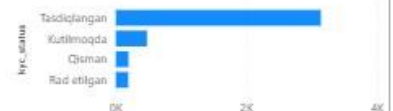
2K

Faol Mijozlar Soni

10bn

Depozit Portfeli

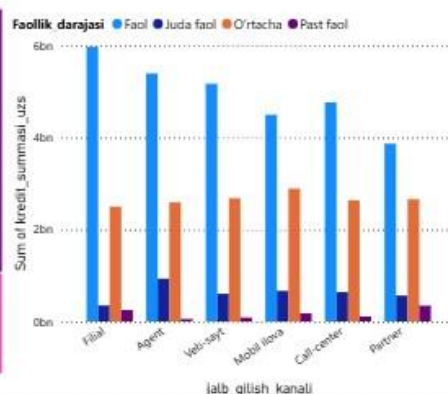
KYC Verifikatsiyasi Holati



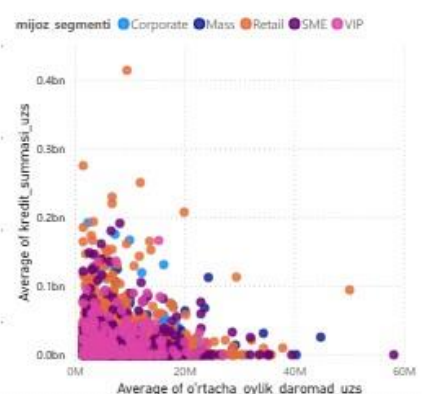
Mijoz Segmentlari va Depozitlar



Jalb Qilish Kanallari va Faollik



Daromad va Kredit Hajmi Korrelyatsiyasi



KREDIT RISKI VA NPL PORTFELI MONITORINGI

3bn

Muddati O'tgan Qarz...

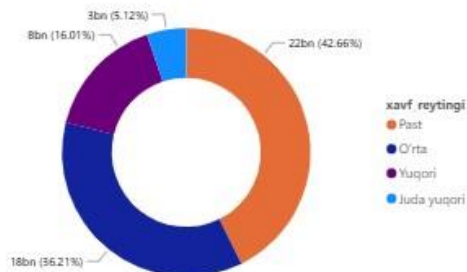
618.25

Average of kredit_reytingi

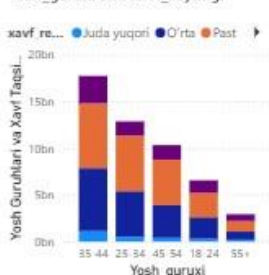
789

Yuqori Riskli Mijozlar

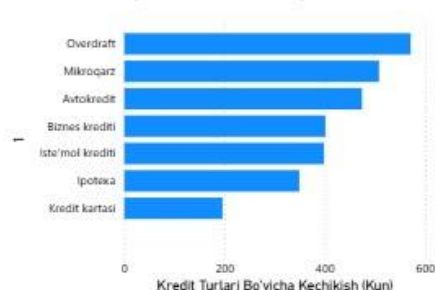
Xavf Darajalari Bo'yicha Kreditlar by xavf_reytingi



Yosh Guruhlari va Xavf Taqsimoti by Yosh_guruxi and xavf_reytingi



Kredit Turlari Bo'yicha Kechikish (Kun) by 1



viloyat	Juda yuqori	O'rta	Past	Yuqori	Total
Andijon	423086560	1969862484	1720195731	369854104	4482998879
Buxoro	169291462	1295427113	1065331152	401770492	2931820219
Farg'ona	149042559	1408027403	1404442028	391613104	3353125094
Jizzax	14618471	588190735	584744034	411405232	1598958472
Namangan	209597231	1151843517	2014134928	705443195	4081018871
Navoiy	0	795366946	387443677	74882839	1257693462
Qashqadaryo	202769464	1174083450	1122045344	785754554	3284652812
Qoraqalpog'iston	0	415151979	477815435	249362080	1142329502
Samarqand	193819256	1927508393	1927888974	1048457278	5097673901
Sirdaryo	5171561	254943426	684612222	163629715	1108356924
Surxondaryo	200261090	708614472	866213928	329100911	2104190401
Toshkent shahri	528380896	3888490785	3266046195	1953765007	11656682883
Toshkent viloyati	314659665	2117510423	3065066351	848682517	6345918956
Xorazm	170246931	563727068	910050017	342000746	1986024762
Total	2580945146	18258748194	21516030016	8075721782	50431445138

BANK MIJOZLARI VA KREDIT PORTFELI TAHLILI

4K

Jami Mijozlar

10bn

Sum of kredit_s...

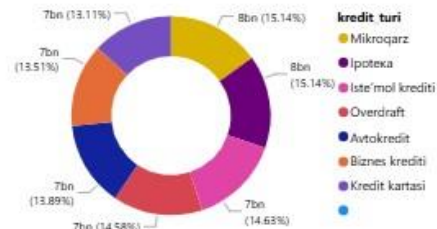
3bn

Muddati O'tgan...

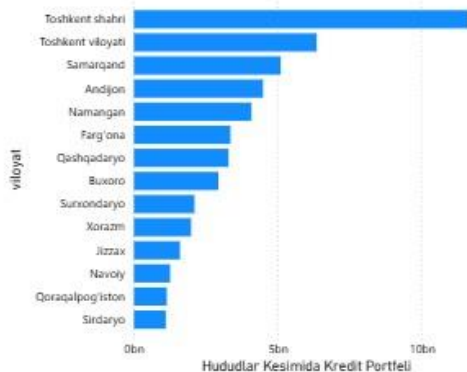
50bn

Sum of kredit_s...

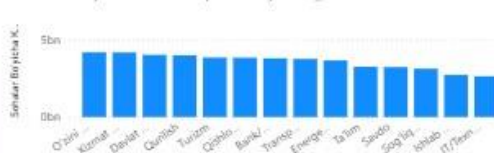
Kredit Turlari Tarkibi by kredit_turi



Hududlar Kesimida Kredit Portfeli by viloyat



Sohalar Bo'yicha Kredit Taqsimoti by kasb_sohasi



- kasb_sohasi
- xavf_reytingi
- ☐ Juda yuqori
 - ☐ O'rta
 - ☐ Past
 - ☐ Yuqori
- jalb qilish kanali
- ☐ Agent
 - ☐ Call-center
 - ☐ Filial
 - ☐ Mobil ilova
 - ☐ Partner
 - ☐ Veb-sayt